FL Simulation Results

File: test_NormClipp_2

Federated Learning Configuration

Total Clients (N): 10

Malicious Clients (M): 3

Neural Network: SimpleNN

Training Rounds: 15

Poisoned Rounds (R): 15

Distribution Strategy: last

Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip

Attack Intensity: 10.0% (0.10)

Poisoned Data Percentage: 100.0% (1.00)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 100.0% (1.00) of data for 15 rounds with 3 malicious clients.

Simulation Results

Init Accuracy: 0.7928

Clean Accuracy: 0.8677

Clean + DP Protection Accuracy: 0.8645

Poisoned Accuracy: 0.5266

Poisoned + DP Protection Accuracy: 0.5540

Drop (Clean - Poisoned): 0.3411

Drop (Clean - Init): 0.0749

Drop (Clean DP - Init): 0.0717

Drop (Poisoned - Init): -0.2662

Drop (Poisoned DP - Init): -0.2388

DP Protection Method: norm_clipping

GPU Used: GPU 2

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8744
Clean Recall: 0.8677
Clean F1 Score: 0.8671

Clean DP Precision: 0.8711
Clean DP Recall: 0.8645
Clean DP F1 Score: 0.8639

Poisoned Precision: 0.6966
Poisoned Recall: 0.5266
Poisoned F1 Score: 0.5431

DP Protection Precision: 0.7047
DP Protection Recall: 0.5540
DP Protection F1 Score: 0.5645

Summary

FL Simulation Complete
Task: 6697e119-0436-4893-9229-014e5632bccc
GPU: GPU 2
Init Accuracy: 0.7928
Clean Accuracy: 0.8677
Clean DP Accuracy: 0.8645
Poisoned Accuracy: 0.5266
Data Poison Protection Accuracy: 0.5540
Drop (Clean - Poisoned): 0.3411
Drop (Clean - Init): 0.0749
Drop (Clean DP - Init): 0.0717
Drop (Poisoned - Init): -0.2662
Drop (Poisoned_DP - Init): -0.2388
Data Poison Protection Method: norm_clipping
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8744
Clean Recall: 0.8677
Clean F1 Score: 0.8671
Clean DP Precision: 0.8711
Clean DP Recall: 0.8645
Clean DP F1 Score: 0.8639
Poisoned Precision: 0.6966
Poisoned Recall: 0.5266
Poisoned F1 Score: 0.5431
DP Protection Precision: 0.7047
DP Protection Recall: 0.5540
DP Protection F1 Score: 0.5645